

RAITE 2026, AI in Education

QuizApp: a quiz from the module itself, every question with its source

A teacher drops in a module and gets a quiz in about 20 seconds, shares it by link or QR code, sees what the class missed, and prints Set A and Set B with the answer key.

The problem

Quizzes take a teacher's evening, and AI help cannot show its work

Written by hand

Each quiz is typed from the module, then retyped as Set A and Set B, with a key and a table of specifications for the department.

Unchecked AI

Chat tools write questions that sound right. Nothing shows which line of the module a question came from, so every one has to be checked.

Unfair checking

Online forms mark "Ductility" wrong when the key says "ductility", and "five" wrong when it says "5". Essays still need a person.

What it does

From a module to a graded, printed test

Teacher

Drop the module

A PDF or photos of the pages. It is read the moment it is picked.

AI

Draft in about 20 s

Four kinds of question, each quoting its sentence. A chat refines it; a picture per question comes from the module's own figures or Wikimedia Commons.

Teacher

Share

A link and a QR code; a name is enough. Shuffled papers, a time limit, results held until released.

Respondents

Answer

Feedback after each answer, with the source sentence.

Teacher

See and print

Most missed, what to teach again, Set A and B, key, table of specifications.

Grounded by design

Every question cites its sentence, and plain code checks it

Quote	The AI copies the sentence that supports each answer, with its page.	Found in science7-metals.pdf, page 1 Not found in the file
Check	No AI: the quote must appear in the file, word for word after normalizing, or 85% of its words in order.	
Flag	A quote not in the file is marked, so the teacher checks that question first.	
Photos and scans	Read once by the AI, word for word, so they can be checked too.	

Exact where it can be, AI where it must, and always labeled

Multiple choice

Graded exactly, in 5 to 11 ms on the server.

True or false

Graded exactly, in English or Filipino (Tama o Mali).

Identification

Case, spacing and number words ("lima" for 5) first; only a near miss goes to the AI, median 3.1 s.

Essay

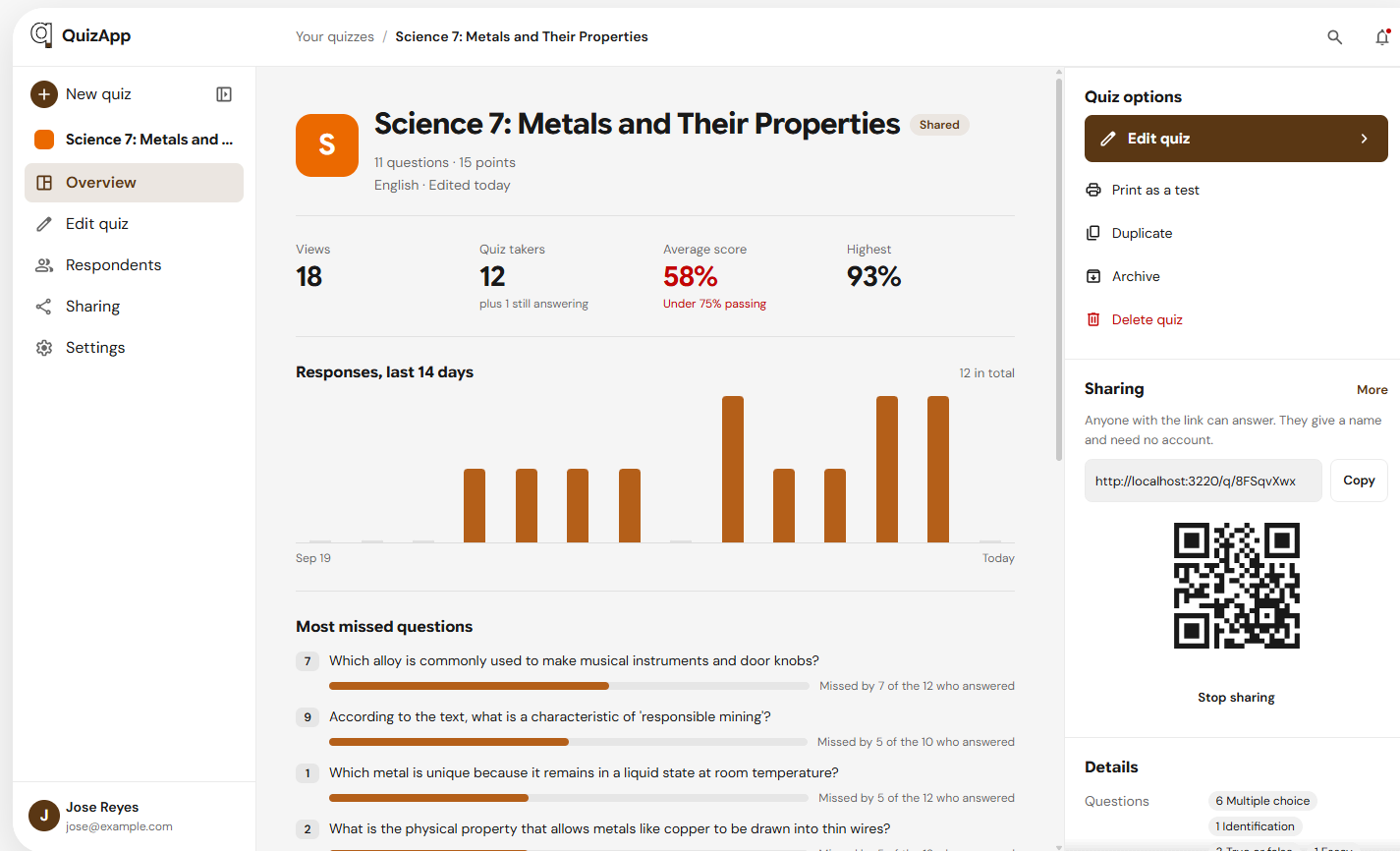
Scored by the AI against the teacher's rubric, 3 to 7.5 s, with a notice that it can be wrong.

The teacher decides

Every AI verdict is labeled "AI", and any score can be changed. Both AI checks can be turned off: then identification is exact and essays wait for the teacher's score.

Results

What the class missed, and what to teach again



- Views, takers, the average against the 75% passing mark, the highest.
- The most missed questions, with how many missed each.
- A note from the AI on what to teach again, written from anonymous counts.
- Reports from respondents, when an answer looks wrong.

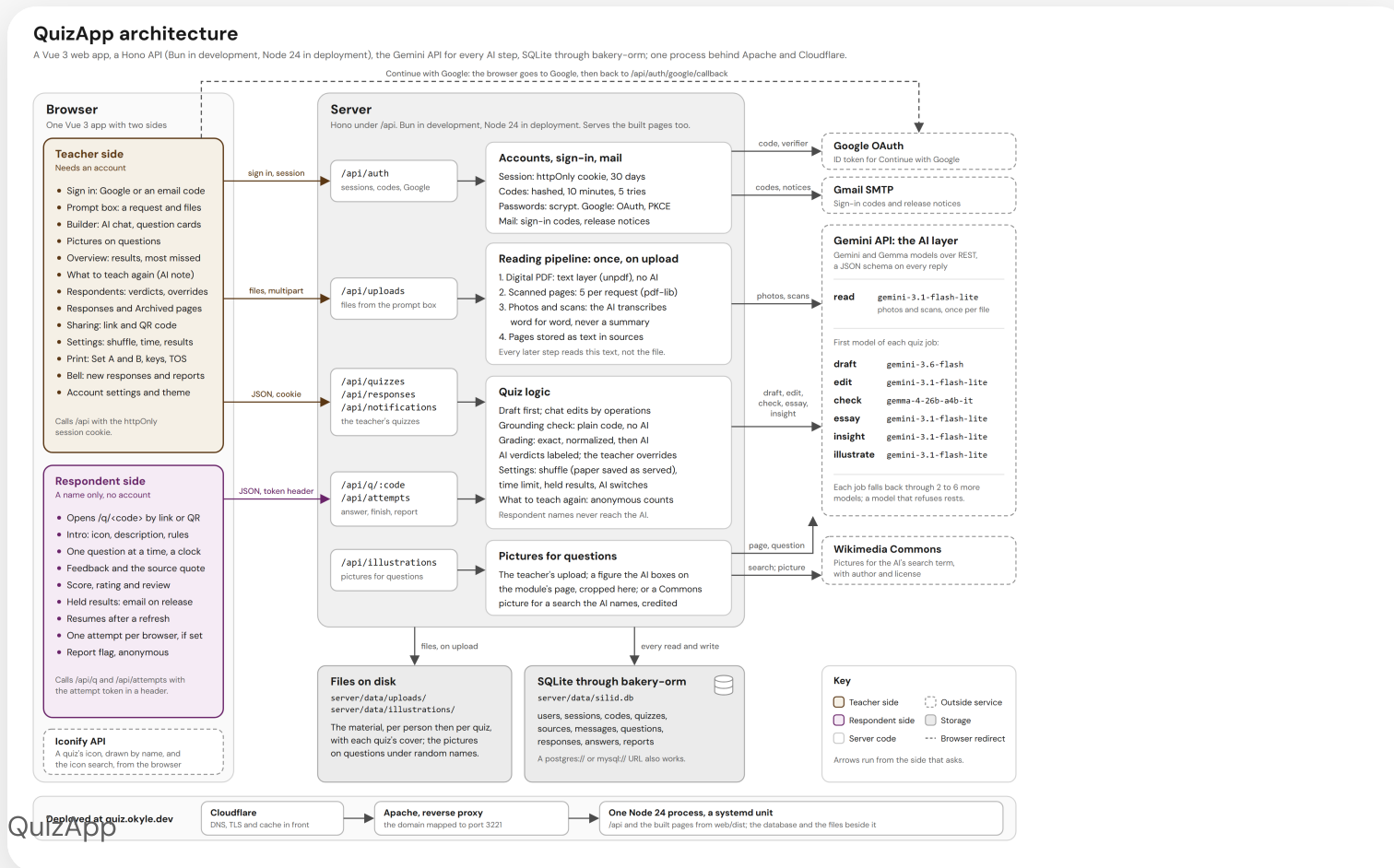
Set A, Set B, the keys and the table of specifications

- Laid out as Philippine test papers are: Name and Score, then Test I to IV with directions.
- Set B moves every question and every option; no multiple-choice letter repeats Set A's at the same number.
- A table of specifications by topic and level of thinking, weighted by points.
- Black and white, on short, long or A4 bond.

How it is built

Vue on the front, Node on the back, Gemini for the AI

- Vue 3.5 and Vite on the front.
- Hono on Node 24, SQLite through bakery-orm; one process serves the API and the pages, behind Apache and Cloudflare at quiz.okyle.dev.
- Gemini and Gemma models through the Gemini API, a chain per job: a model that refuses rests and the next takes over.



Nothing hidden from the teacher, nothing asked of the student

For respondents

A name and no account. Answers never reach the browser before a question is answered. Reports are anonymous.

For teachers

Every AI verdict is labeled and can be changed. Every quote is checked against the file. The AI's note reads counts, not names.

Measured on this build

Fast enough for a class, checked enough for a grade

17 to 21 s

A draft of 10 to 12 questions from a module.

5 to 11 ms

Grading a multiple-choice answer on the server.

3.1 s

The median AI check of a near-miss identification answer.

5 to 6.4 s

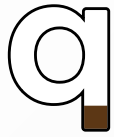
A chat edit to the quiz.

6.7 s

Reading a photo of a page, word for word.

33 s

Reading a 21-page scanned PDF.



Thank you

QuizApp, by Kyle Cyrus Obille, for RAITE 2026.